

Ruiyang Zhou

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📍 Atlanta, GA

Education

Georgia Institute of Technology

M.S. in Computational Science and Engineering (GPA: 3.87 / 4.00)

August 2025 - May 2027

Atlanta, GA

- **Focus:** Distributed systems, high-performance computing, and machine learning systems

University of Virginia

B.A. in Computer Science with Highest Distinction (GPA: 3.87 / 4.00)

August 2020 - May 2025

Charlottesville, VA

Technical Skills

Languages: Python, Java, TypeScript, C/C++, SQL, Shell

Software Engineering: Spring Boot, PostgreSQL, Redis, Kafka, Google Cloud Platform, Docker, Git, RESTful API, Express, Prisma, Django, MyBatis, Next.js, React.js, Electron, DB2, Linux, CI/CD, Prometheus, asyncio, Slurm, OpenMPI, OpenCL

AI/ML: PyTorch, Hugging Face, vLLM, CUDA, ChromaDB, PaddleOCR

Publications

G. Mao, **R. Zhou**, "Energy Characterization of KV Cache Offloading Under Agentic Workloads," *HotCarbon 2026: Workshop on Sustainable Computer Systems* (to appear in *ACM SIGEnergy Energy Informatics Review*, 6(2)).

Experience

Co-Founder & Software Engineer

Orivya (Georgia Tech CREATE-X Startup Launch)

May 2026 - July 2026

Atlanta, Georgia

- Led the 0-to-1 development of a stock footage analysis and licensing platform for video creators with an **Express/Prisma** backend on **PostgreSQL**, a **Next.js** dashboard, and an **Electron** desktop client, with the API and workers deployed on **GCP Cloud Run**.
- Implemented recoverable async jobs for AI tagging using PostgreSQL SKIP LOCKED and expiring leases; workers stage videos in a private **Cloud Storage** bucket, call **Gemini 2.5 Flash** via **Vertex AI** with response schemas, and validate output with **Zod**. Added **FFmpeg** and **Cloud Vision** Web Detection screening to flag high-confidence findings and route uncertain cases to human review.
- Designed a checkout flow with per-listing row locks to enforce license limits and prevent duplicate orders, **Redis** Lua scripts for atomic reservation checks, and a transactional outbox publishing order events to **Kafka** for idempotent earnings processing. Load-tested at **200** concurrent clients: **~1.2K orders/s** sustained, **178 ms** p95, zero errors.

Software Engineer Intern

China CITIC Bank

June 2025 - August 2025

Chongqing, China

- Worked on rebuilding a legacy client account management backend, developing **RESTful APIs** in **Java/Spring Boot**, mapping them to database queries with **MyBatis**, and integrating with the existing **DB2** database.
- Implemented Chinese Pinyin name fuzzy search in the management system using Apache string-similarity and indexing libraries, improving typo tolerance and reducing failed searches by **33%**.
- Integrated an RPA script into the complaint board to download data regularly, and built a **Python automation** pipeline that processes KPI data (handling time, complaint volume, active departments) with **Pandas** and generates Word reports, reducing reporting time from 3–4 hours to **~30 seconds**.

FPGA Research Assistant

University of Virginia

December 2023 - May 2025

Charlottesville, Virginia

- Built on an AMD Alveo U280 **FPGA** with **Xilinx Vitis (C++)**, extended a VM-based regular expression matching engine kernel from single-pattern to single-input, multi-pattern matching.
- Parallelized the engine across **13** concurrent kernels with **OpenCL**, raising throughput from 240 MB/s on a single kernel to **~1.5 GB/s** (**~6×**); fixed kernel deadlocks and validated multi-kernel sync with HLS stream guards.
- Evaluated on the **AutomataZoo** benchmark suite, converting its regular expression formats into the form accepted by the regex virtual machine.

Projects

Policy-Aware Expense Reporting Agent | Python, FastAPI, ChromaDB, PaddleOCR, Redis, Docker

- Built an expense reporting agent combining LLM, embedding, and pattern-based intent recognition to handle pre-expense inquiries and submissions, matching requests to applicable policies and spending limits.
- Implemented a policy **RAG** pipeline on **ChromaDB** with LLM query rewriting into three sub-queries, parallel retrieval, deduplication and re-ranking, reaching **88.7%** answer accuracy and **91.2%** retrieval recall on an internal evaluation set; external calls are guarded by caching, timeouts and circuit-breaker fallback.
- Integrated **PaddleOCR** and structured extraction to populate reimbursement forms from invoices, cross-checking expense details against pre-expense records and applicable policies; reached **89.4%** form accuracy across **180 end-to-end test cases**.